Chapter 30

Trig identities and equations

always true, or only sometimes?

ALWAYS TRUE

$$\sin^2\theta + \cos^2\theta = 1$$

try any angle you like

SOMETIMES TRUE

$$\sin\theta = 0.5$$

only at 30°, 150°, 390°...

Some trig statements hold for **every** angle: these are **identities**, tools you can swap in whenever convenient. Others hold only for **particular** angles: these are **equations**, puzzles to solve.

The big idea for equations: trig waves repeat, so an equation usually has **many** answers, and your calculator only tells you one of them. We'll catch them all with a picture.

Goes with **Lecture 6** ($\cos^2 + \sin^2 = 1$, multiple values, $r\sin(\theta + \alpha)$, the advanced question).

∿ sin² + cos² = 1 is Pythagoras on the circle

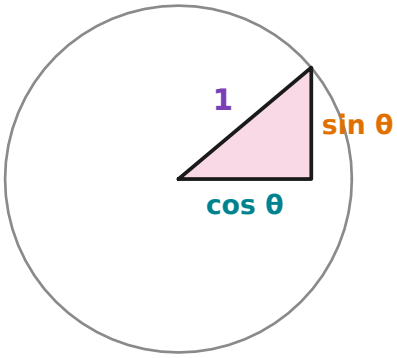
RULE: the Pythagorean identity

For **every** angle: $\sin^2\theta + \cos^2\theta = 1$. Rearranged: $\cos^2\theta = 1 - \sin^2\theta$ and $\sin^2\theta = 1 - \cos^2\theta$.

Also $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, and dividing by $\cos^2\theta$: $\tan^2\theta + 1 = \dfrac{1}{\cos^2\theta}$.

$\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$.

On the unit circle (Chapter 29) the point at angle θ is $(\cos\theta, \sin\theta)$, and every point on it obeys $x^2 + y^2 = 1$. Swap in $x = \cos\theta$, $y = \sin\theta$:



$$(\cos\theta)^2 + (\sin\theta)^2 = 1^2$$

$$\cos^2\theta + \sin^2\theta = 1$$

short sides squared, added = hypotenuse squared

Notation: $\sin^2\theta$ means $(\sin\theta)^2$: work out the sine, then square it. It does **not** mean $\sin(\theta^2)$. (Lecture 6 writes "cos 25² + sin 25² = 1", which you should read as $\cos^2 25° + \sin^2 25°$.)

θ = 25°

(0.9063)² + (0.4226)²

0.8214 + 0.1786 = 1 ✓

θ = 550°

(-0.9848)² + (-0.1736)²

0.9698 + 0.0302 = 1 ✓

θ = -73°

(0.2924)² + (-0.9563)²

0.0855 + 0.9145 = 1 ✓

Using it: if $\sin\theta = \frac{3}{5}$ (and θ is acute), then $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos\theta = \frac{4}{5}$. No angle needed. (It's the 3-4-5 triangle in disguise.)

Dividing the identity by $\cos^2\theta$ also shows $\tan\theta = \frac{\sin\theta}{\cos\theta}$ fits in: $\tan^2\theta + 1 = \frac{1}{\cos^2\theta}$.

PROCEDURE: given one ratio, find the others

1. Use the identity: $\cos^2\theta = 1 - \sin^2\theta$ (or $\sin^2\theta = 1 - \cos^2\theta$).
2. Square-root. This gives **two** possible values, $\pm$.
3. Choose the sign from the **quadrant** θ is in (CAST, Chapter 29): in Q2 only sin is positive, in Q3 only tan, in Q4 only cos.
4. Then $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, keeping the signs.

Example: $\sin\theta = \frac{3}{5}$ and θ is **obtuse** (between $90°$ and $180°$). Find $\cos\theta$ and $\tan\theta$.

$$\cos^2\theta = 1 - \left(\tfrac{3}{5}\right)^2 = \tfrac{16}{25}$$ ← step 1

$$\cos\theta = \pm\tfrac{4}{5}$$ ← step 2: two candidates

$$\cos\theta = -\tfrac{4}{5}$$ ← step 3: obtuse means Q2, where cos is negative

$$\tan\theta = \frac{3/5}{-4/5} = -\tfrac{3}{4}$$ ← step 4

Check: $\theta = 180° - 36.87° = 143.13°$, and a calculator gives $\cos 143.13° = -0.8$ ✓.

Solving: catch every crossing

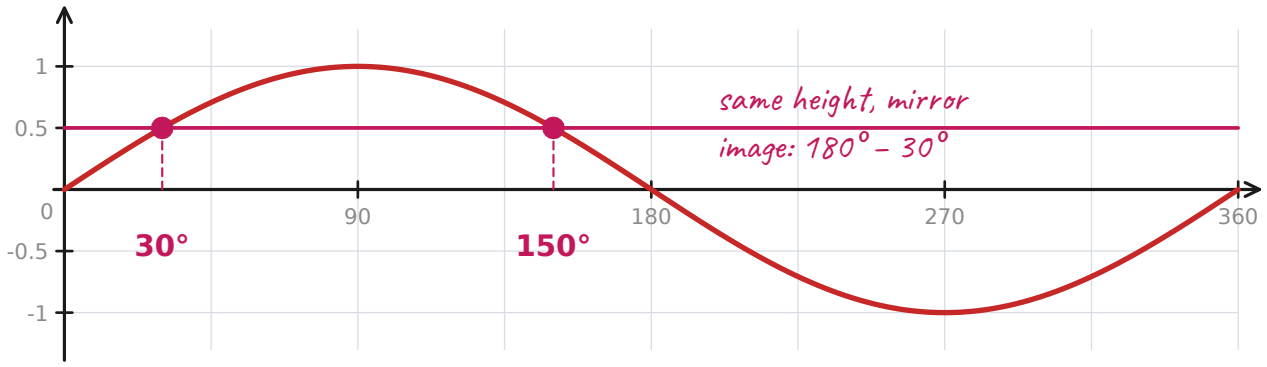
RULE: trig equations have many solutions

The calculator ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) gives **one** answer, the principal value.

Each function has a **partner rule** giving a second answer in the same turn (table below).

Every answer repeats every whole turn: add or subtract $360°$ (2π) to list all the solutions in the range asked for.

Solve $\sin\theta = 0.5$ for $0° \le \theta \le 360°$. The calculator's $\sin^{-1}(0.5)$ says 30°. But the Ferris-wheel seat is at height 0.5 **twice** per turn: once going up, once coming down. Draw the line $y = 0.5$ across the wave:



The second answer comes from the wave's symmetry. Each function has its own partner rule. If the calculator's answer is θ_1:

Equation	Calculator gives θ₁ in	Partner (degrees)	Partner (radians)	Why
$\sin\theta = k$	$-90°$ to $90°$	$180° - \theta_1$	$\pi - \theta_1$	wave symmetric about $90°$
$\cos\theta = k$	$0°$ to $180°$	$360° - \theta_1$	$2\pi - \theta_1$	wave symmetric about $180°$
$\tan\theta = k$	$-90°$ to $90°$	$\theta_1 + 180°$	$\theta_1 + \pi$	tan repeats every $180°$

Then add or subtract **whole turns** (360°) to reach any other range. So $\sin\theta = 0.5$ also has $390°, 510°, -210°, \ldots$

Lecture 6's example: $\cos^{-1}(-1)$. The calculator says 180° (π). But cos touches -1 at the bottom of **every** dip: $\theta = \pi, 3\pi, -\pi, -3\pi, \ldots$, i.e. $180°, 540°, -180°, -540°$.

A harder one, step by step

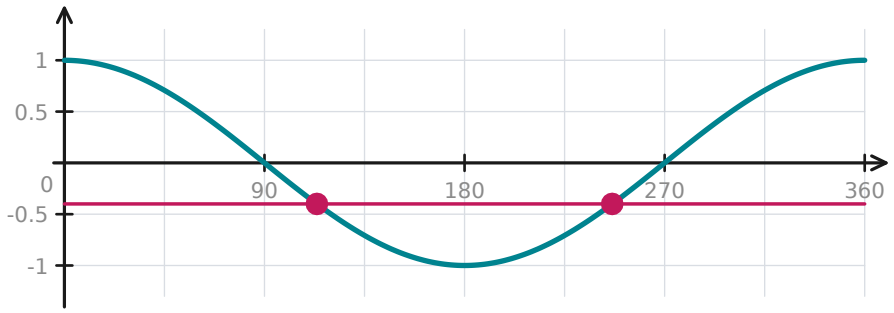
Solve $5\cos\theta + 2 = 0$ **for** $0° \le \theta \le 360°$**.**

$$5\cos\theta = -2$$ ← − 2 both sides

$$\cos\theta = -0.4$$ ← ÷ 5: get the cos on its own first

$$\theta = \cos^{-1}(-0.4) = 113.58°$$ ← calculator (one answer)

$$\theta = 360° - 113.58° = 246.42°$$ ← cos partner rule



two crossings, mirror images about 180°

Check: $5\cos 246.42° + 2 = 5(-0.4) + 2 = 0$ ✓.

When the calculator gives a negative angle

PROCEDURE: solving sin θ = k, cos θ = k or tan θ = k in a range

1. Rearrange so the trig function is on its own: $\sin\theta = k$.
2. Get the principal value $\theta_1 = \sin^{-1} k$ (or $\cos^{-1} k$, $\tan^{-1} k$). It may be **negative** or out of range: keep it anyway.
3. Get the partner: $180° - \theta_1$ for sin, $360° - \theta_1$ for cos, $\theta_1 + 180°$ for tan (radians: $\pi - \theta_1$, $2\pi - \theta_1$, $\theta_1 + \pi$).
4. Add or subtract $360°$ (2π) to each answer until it lands in the range. Keep every one that does.
5. Check each answer on the calculator.

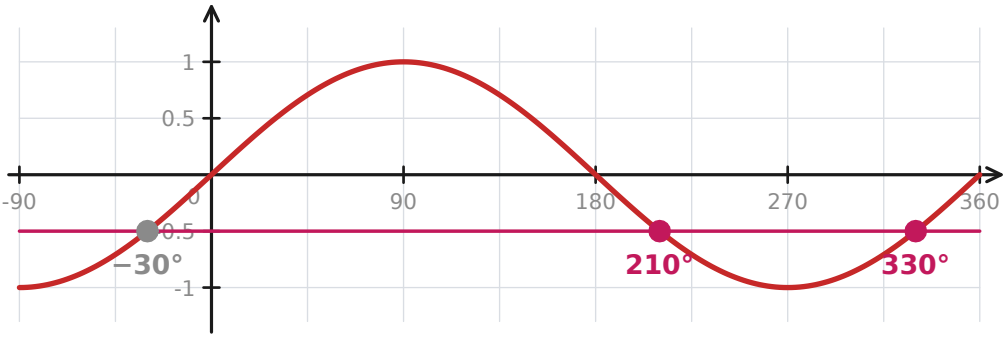
Example 1: solve $\sin\theta = -0.5$ for $0° \le \theta < 360°$.

$$\theta_1 = \sin^{-1}(-0.5) = -30°$$ ← step 2: out of range, but keep it

$$180° - (-30°) = 210°$$ ← step 3: sin partner

$$-30° + 360° = 330°$$ ← step 4: bring θ_1 into range

$$\theta = 210°,\ 330°$$



calculator: −30° (grey), outside the range.

In range: 210° and 330°

Example 2: solve $\tan\theta = -2$ for $0° \le \theta < 360°$.

$$\theta_1 = \tan^{-1}(-2) = -63.43°$$

$$-63.43° + 180° = 116.57°, \quad -63.43° + 360° = 296.57°$$

tan: add 180° each time

Example 3 (radians): solve $\sin\theta = -0.4$ for $0 \le \theta < 2\pi$.

$$\theta_1 = \sin^{-1}(-0.4) = -0.412$$ ← RAD mode

$$\pi - (-0.412) = 3.553, \qquad -0.412 + 2\pi = 5.872$$

partner π − θ₁; add 2π

Multiple angles: sin 2θ = 0.5

PROCEDURE: equations like sin(nθ) = k

1. Call the whole angle $\phi = n\theta$. Multiply the range by n too: if $0° \le \theta < 360°$ then $0° \le \phi < n \times 360°$.
2. Solve for ϕ in the stretched range: principal value, partner, then add $360°$ to each until you pass the top of the range.
3. Divide every ϕ by n to get θ. Expect n times as many answers.

Example: solve $\sin 2\theta = 0.5$ for $0° \le \theta < 360°$.

$$\phi = 2\theta, \quad 0° \le \phi < 720°$$ ← *step 1: double the range*

$$\phi = 30°,\ 150°,\ 390°,\ 510°$$ ← *30°, its partner 150°, then + 360°*

$$\theta = 15°,\ 75°,\ 195°,\ 255°$$ ← *step 3: halve*

Four answers, not two: $\sin 2\theta$ has period $180°$ (Chapter 29), so it goes through two full waves between $0°$ and $360°$.

✗ Common mistakes

✗ $\sin\theta = -0.5 \Rightarrow \theta = -30°$ ✓ $\theta = 210°,\ 330°$

−30° is outside 0° to 360°: use the partner and add 360°

✗ $\sin 2\theta = 0.5 \Rightarrow \theta = 15°, 75°$ ✓ $\theta = 15°, 75°, 195°, 255°$

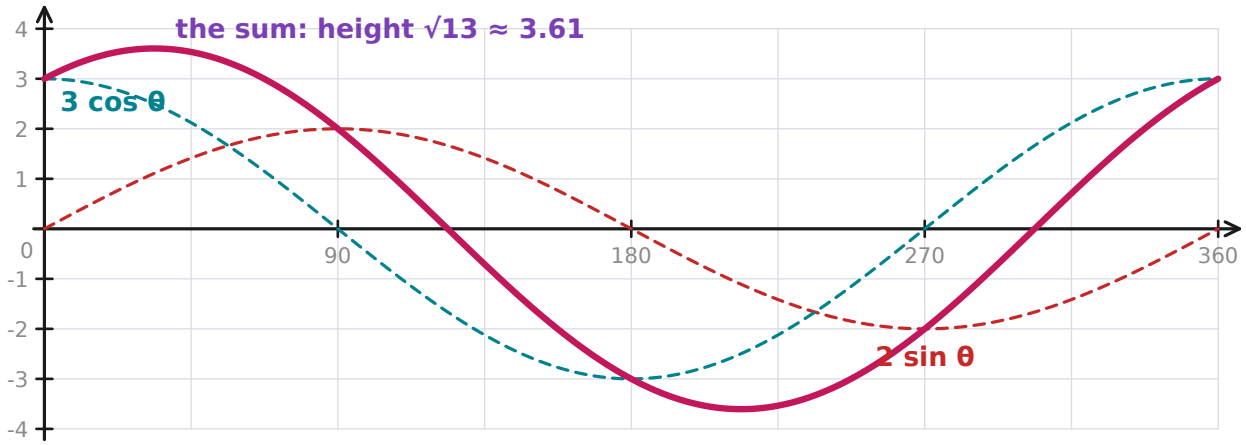
stretch the range to 0°–720° before halving

Two waves make one: r sin(θ + α)

PROCEDURE: a sin θ + b cos θ = r sin(θ + α)

1. $r = \sqrt{a^2 + b^2}$.
2. $\alpha = \tan^{-1}\dfrac{b}{a}$, then the quadrant fix: sketch the point (a, b); if $a < 0$, add $180°$.
3. The largest value is r, the smallest $-r$.
4. To solve $a\sin\theta + b\cos\theta = c$: write $\sin(\theta + \alpha) = \frac{c}{r}$, solve for the whole bracket $\phi = \theta + \alpha$ (shift the range by α), then subtract α.

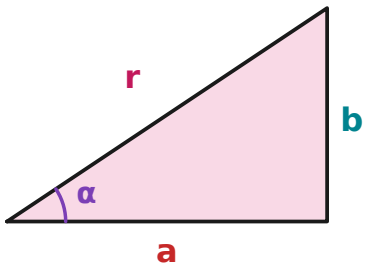
Add a sine wave and a cosine wave, $2\sin\theta + 3\cos\theta$, and something lovely happens: the result is **a single sine wave**, just taller and shifted:



Lecture 6 writes $a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$, with

$$r = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1}\frac{b}{a}$$



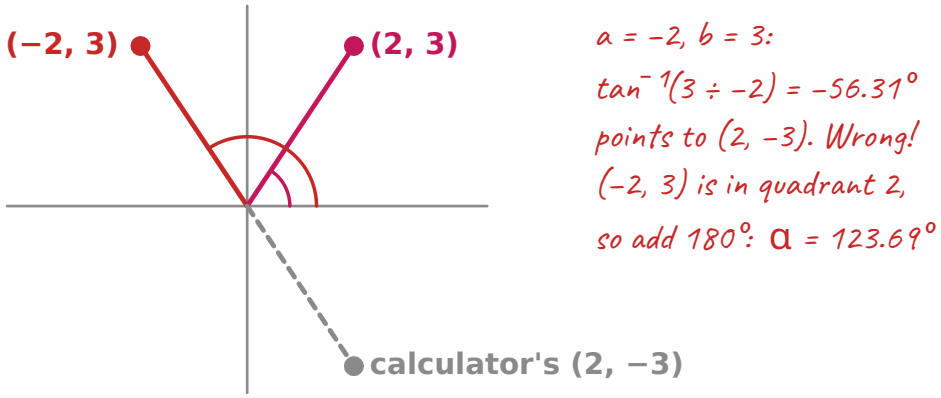
a and b as the sides of a right triangle

Why? There's a standard identity for the sine of a sum (the **compound angle formula**):

$$\sin(\theta + \alpha) = \sin\theta\cos\alpha + \cos\theta\sin\alpha$$

Multiply by r and match with $a\sin\theta + b\cos\theta$: we need $r\cos\alpha = a$ and $r\sin\alpha = b$. Those are exactly the sides of the triangle above (CAH and SOH with hypotenuse r), and Pythagoras gives r. Lecture 6 notes this is the same move as going from (x, y) to polar coordinates (Chapter 33).

Watch the signs. $r\cos\alpha = a$ and $r\sin\alpha = b$ say that α is the angle of the **point** (a, b). When a and b are both positive that point is in quadrant 1 and $\tan^{-1}\frac{b}{a}$ is right. When a is **negative**, the calculator still answers in the right-hand half, so it points the wrong way:



So: **sketch the point** (a, b) **first**. If $a < 0$ (quadrants 2 and 3), add $180°$ to the calculator's answer. Chapter 33 builds this **quadrant fix** properly for polar coordinates; it's exactly the same job.

Lecture 6's advanced question, both answers

Solve $2\sin\theta + 3\cos\theta = 2$ **for** $0° \le \theta \le 360°$**.**

$$r = \sqrt{2^2 + 3^2} = \sqrt{13}, \quad \alpha = \tan^{-1}\tfrac{3}{2} = 56.31°$$

a = 2, b = 3

$$\sqrt{13}\,\sin(\theta + 56.31°) = 2 \leftarrow \text{one wave now}$$

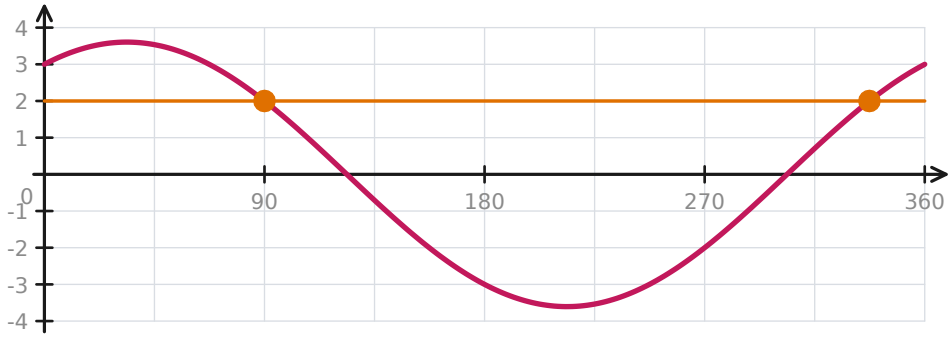
$$\sin(\theta + 56.31°) = \frac{2}{\sqrt{13}} = 0.5547 \leftarrow \div\sqrt{13}$$

Treat the whole bracket as one angle, call it ϕ ("fie"). The calculator gives $\phi = 33.69°$; the sin partner is $180° - 33.69° = 146.31°$:

$$\theta + 56.31 = 33.69 \;\Rightarrow\; \theta = -22.62° \to 337.38°$$

+ 360° to get into range

$$\theta + 56.31 = 146.31 \;\Rightarrow\; \theta = 90° \leftarrow \text{the one the slide asks you to find}$$



the line y = 2 crosses the wave at 90° and 337.38°

Check 90°: $2\sin 90° + 3\cos 90° = 2(1) + 3(0) = 2$ ✓. Easy to miss, which is exactly why the lecture asks.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$ also $\cos^2\theta = 1 - \sin^2\theta$, $\sin^2\theta = 1 - \cos^2\theta$
tan	$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, $\qquad \tan^2\theta + 1 = \dfrac{1}{\cos^2\theta}$
Choosing ±	after a square root, take the sign from the quadrant (CAST)
Partner: sin	$\theta_1, \quad 180° - \theta_1 \quad (\pi - \theta_1)$ $\sin^{-1}$ gives $-90°$ to $90°$
Partner: cos	$\theta_1, \quad 360° - \theta_1 \quad (2\pi - \theta_1)$ $\cos^{-1}$ gives $0°$ to $180°$
Partner: tan	$\theta_1, \quad \theta_1 + 180° \quad (\theta_1 + \pi)$ $\tan^{-1}$ gives $-90°$ to $90°$
All solutions	add or subtract $360°$ (2π) to each; keep those in range, even if θ_1 itself is negative
Multiple angles	for $\sin n\theta = k$: multiply the range by n, solve, divide by n
r sin form	$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$ $r = \sqrt{a^2 + b^2}$, $\alpha = \tan^{-1}\frac{b}{a}$ (+180° if $a < 0$); max r, min $-r$
Compound angle	$\sin(A + B) = \sin A\cos B + \cos A\sin B$

Your turn

Range 0° to 360° unless stated. Answers in grey.

Identities

1. If $\cos\theta = 5/13$ (θ acute), find $\sin\theta$
2. Simplify $3\sin^2 x + 3\cos^2 x$
3. Is $\sin(2\theta) = 2\sin\theta$ an identity? Test $\theta = 90°$.

1. 12/13 2. 3 3. no: sin 180° = 0 but 2 sin 90° = 2

Equations

4. $\sin\theta = 0.8$
5. $\cos\theta = 0.5$
6. $\tan\theta = 1$
7. $2\sin\theta + 1 = 0$

4. 53.13°, 126.87° 5. 60°, 300° 6. 45°, 225° 7. 210°, 330°

r sin form

8. Write $3\sin\theta + 4\cos\theta$ as $r\sin(\theta + \alpha)$
9. Hence the largest value of $3\sin\theta + 4\cos\theta$
10. Solve $3\sin\theta + 4\cos\theta = 5$

8. $5\sin(\theta + 53.13°)$ 9. 5 10. $\theta + 53.13 = 90$, so $\theta = 36.87°$

Find the unknown

11. $\sin\theta = -0.6$
12. $\cos\theta = -0.25$ for $0 \le \theta < 2\pi$ (radians, 3 d.p.)
13. $\tan\theta = -1.5$
14. $\cos 2\theta = 0.5$
15. $\cos\theta = -5/13$ and θ is in quadrant 3. Find $\sin\theta$ and $\tan\theta$.

11. $\theta_1 = -36.87°$; $180° - (-36.87°) = 216.87°$, $-36.87° + 360° = 323.13°$ 12. $\theta_1 = \cos^{-1}(-0.25) = 1.823$; $2\pi - 1.823 = 4.460$ 13. $\theta_1 = -56.31°$; + 180° = 123.69°, + 360° = 303.69° 14. $2\theta = 60°, 300°, 420°, 660°$ (range 0°–720°), so $\theta = 30°, 150°, 210°, 330°$ 15. $\sin^2\theta = 1 - 25/169 = 144/169$, $\sin\theta = -12/13$ (Q3); $\tan\theta = 12/5$